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### Visual media management for devices

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#### Abstract

A server includes a processor programmed to: acquire first metadata of a first media file recorded by a first mobile device; acquire second metadata of a second media file recorded by a second mobile device; determine that the first media file and the second media file are likely recordings of the same event when a similarity exceeds a first threshold. The processor is further programmed to, when the first media file and the second media file are likely recordings of the same event: determine, based on a comparison between the first media file and the second media file, which of the first media file and the second media file is a higher quality recording of the same event; and when the first media file is the higher quality recording, send a link to the first media file to the second mobile device.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This is a Continuation Application of application Ser. No. 18/182,540 filed Mar. 13, 2023, which is a Continuation Application of application Ser. No. 17/567,501 filed Jan. 3, 2022. The disclosures of the prior applications are hereby incorporated by reference herein in their entirety.

### BACKGROUND

(1) Smartphones have become ubiquitous in many areas of the world. In most urban areas, smartphones are carried by almost every adult. The camera (i.e. photo and video recording) technology of smartphones has improved dramatically in recent years. This fact, combined with people's innate desire to take photographs, has made photography with smartphones extremely popular.

(2) The photographic capabilities, memory storage, and constant availability of smartphones have allowed people to document events to an extent never before possible. However, there are still drawbacks, and possibilities for improvement.

(3) For one thing, the size constraints (particularly thickness) of the smartphone place meaningful limits on the zoom capabilities of the smartphone camera. Smartphone manufacturers have made great strides in this area by including arrays of several different lenses with different focal lengths, and using artificial intelligence (AI) to enhance the quality of the photos. AI has been used particularly in the areas of reducing blur for shaky photos, moving objects, or photos of very distant objects. Sound quality for video recordings has also been improved by using arrays of multiple microphones, noise cancellation, and AI enhancement.

(4) However, this does not solve all of the potential problems with recording distant objects or events. First, even with the above enhancements, distant photos are still lower quality than near photos. Second, there could be an intervening (i.e. view blocking) object between the distant object or event being recorded, and the user. Third, for video recordings, the sound quality is greatly reduced for recording distant objects and events. This is true even in view of the above-noted improvements in sound recording.

(5) Thus far, innovators in this field have focused on improving the image/sound recording and processing capabilities of the smartphone itself to address these problems. However, there is a limit to what this approach can achieve.

### SUMMARY

(6) One aspect of the disclosure is a server that includes a processor. The processor is programmed to acquire first metadata of a first media file recorded by a first mobile device. The processor is further programmed to acquire second metadata of a second media file recorded by a second mobile device. The processor is further programmed to determine a similarity between the first media file and the second media file based on a comparison between the first metadata and the second metadata, and determine that the first media file and the second media file are likely recordings of the same event when the similarity exceeds a first threshold. The processor is further programmed to, when the first media file and the second media file are likely recordings of the same event: acquire the first media file and the second media file; determine, based on a comparison between the first media file and the second media file, which of the first media file and the second media file is a higher quality recording of the same event; when the first media file is the higher quality recording, send a link to the first media file to the second mobile device; and when the second media file is the higher quality recording, send a link to the second media file to

the first mobile device.

- (7) In some embodiments, the first metadata and the second metadata include location and time information of the first media file and the second media file, respectively, and the processor is programmed to determine that the similarity exceeds the first threshold when the location and time information of the first media file and the second media file match.
- (8) In some embodiments, the processor is programmed to determine that the location and time information of the first media file and the second media file match when recording times of the first and second media files overlap, and locations of recording the first media file and second media file are less than a distance threshold.
- (9) In some embodiments, the processor is programmed to, after acquiring the first media file and the second media file, confirm that the first media file and the second media file are recordings of the same event by: determining a first subject of the first media file and a second subject of the second media file, by analyzing center regions of frames within the first media file and the second media file; and comparing the first subject and the second subject to determine if they are the same subject.
- (10) In some embodiments, the processor is programmed to compare the first subject and the second subject to determine if they are the same subject by: assigning keypoints to the first subject; locating the keypoints on the second subject; determining a number of the keypoints that are locatable on the second subject; and when the number of locatable keypoints exceeds a threshold, determining that the first subject and the second subject are the same subject.
- (11) In some embodiments, the processor is programmed to determine which of the first media file and the second media file is the higher quality recording of the same event by: calculating a pixel distance between the keypoints in the first media file the keypoints in the second media file; when the pixel distance is higher in the first media file, determine that the first media file is the higher quality recording; and when the pixel distance is higher in the second media file, determine that the second media file is the higher quality recording.
- (12) In some embodiments, the processor is programmed to determine which of the first media file and the second media file is the higher quality recording of the same event by: assigning a first bounding box to the first subject; assigning a second bounding box to the second subject; when the first bounding box is larger than the second bounding box, determine that the first media file is the higher quality recording; and when the second bounding box is larger than the first bounding box, determine that the second media file is the higher quality recording.
- (13) In some embodiments, the processor is programmed to analyze center regions of frames within the first media file and the second media file to determine the first subject of the first media file and the second subject of the second media file, by: identifying objects in the center regions of the frames of the first media file and the second media file; determining that the identified objects are the first subject if they are in the center regions of the frames of the first media file for a majority of a recording time of the first media file; and determining that the identified objects are the second subject if they are in the center regions of the frames of the second media file for a majority of a recording time of the second media file.
- (14) In some embodiments, the processor is programmed to, after acquiring the first media file and the second media file, confirm that the first media file and the second media file are recordings of the same event by: comparing audio recordings in the first media file and the second media file at corresponding time points; and when a similarity in the audio recordings exceeds a threshold, confirming that the first media file and the second media file are recordings of the same event.
- (15) In some embodiments, the first metadata includes a description of a first subject of the first media file, and the second metadata includes a description of a second subject of the second media file and the processor is programmed to determine that the similarity exceeds the first threshold when the descriptions of the first subject and the second subject match.
- (16) One aspect of the disclosure is a mobile device that includes a camera configured to generate

an image recording. The mobile device further includes a memory configured to store the image recording and metadata of the image recording. The mobile device further includes a communicator configured to communicate on a network, and a display. The mobile device further includes a processor programmed to: instruct the communicator to send the metadata of the image recording to a networked device via the network; receive an instruction from the networked device to send the image recording to the networked device, based on a determination that the image recording and another image recording recorded by another mobile device are likely recordings of the same event because a similarity between the metadata and metadata of the other image recording exceeds a first threshold; instruct the communicator to send the image recording to the networked device; receive a link to the other image recording from the networked device via the communicator, based on a determination that the other image recording is higher quality than the image recording; and instruct the display to display the link in association with the image recording.

(17) In some embodiments, the metadata includes location and time information of the image recording. In some embodiments, the processor is programmed to: determine a subject of the image recording by analyzing center regions of frames within the image recording; identify the subject of the image recording; and include a description of the subject of the image recording in the metadata. In some embodiments, the processor is programmed to: assign a bounding box to the subject of the image recording; and include location information of the bounding box in the metadata. In some embodiments, the processor is programmed to: assign keypoints to the subject of the image recording; and include data of the keypoints in the metadata. In some embodiments, the processor is programmed to analyze the center regions of frames within the first image recording to determine the subject of the image recording, by: identifying objects in the center regions of the frames of the image recording; and determining that the identified objects are the subject if they are in center regions of the frames of the image recording for a majority of a recording time of the image recording.

(18) In some embodiments, the processor is programmed to: receive a second link to a second image recording from the networked device via the communicator, based on a determination that the second image recording is higher quality than the image recording; and instruct the display to display the link and the second link in association with the image recording. In some embodiments, the processor is programmed to: receive a user selection of the image recording from the display; and selectively display the link in response to the user selection of the image recording. In some embodiments, the processor is programmed to: receive a user selection of the image recording from the display; and selectively display the link and the second link in response to the user selection of the image recording. In some embodiments, the processor is programmed to: receive a user selection of the link from the display; and instruct the communicator to download the other image recording in response to the user selection of the link.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of a system including a server according to one embodiment.
- (2) FIG. 2 is an illustration of an event being recorded using two mobile devices according to one embodiment.
- (3) FIG. 3 is an illustration of two mobile devices playing their respective recordings of the same event.
- (4) FIG. 4 is an illustration of a mobile device playing its recording of the event.
- (5) FIG. 5 is an illustration of a user interface for viewing media files according to one embodiment.

### DETAILED DESCRIPTION OF EMBODIMENTS

(6) FIG. 1 is a block diagram of a system **100** including a server **110** according to one embodiment. The system **100** includes at least two devices (i.e. first device **120** and second device **140**). These devices are computerized mobile devices such as smartphones, tablets, phablets, glasses, head-mounted devices, wrist-mounted devices, clothing-mounted devices, etc. . . . Server **110** communicates with first device **120** and second device **140** via network **130**. Network **130** may be a public network such as the internet, or a private network such as a local wide network. In some embodiments, network **130** is excluded altogether and first device **120** and second device **140** (and optionally server **110**) communicate using a two-way networking protocol such as Bluetooth.

(7) Server **110** includes a processor **112**, memory **116**, and communicator **114**. First device **120** includes a processor **122**, memory **126**, and communicator **124**. Second device **140** includes a processor **142**, memory **146**, and communicator **144**. The communicators **114**, **124**, and **144** can be e.g. network communication hardware. These components are used to perform the functions of recording real events and sharing the recordings to improve mobile device user experiences.

(8) For example, FIG. 2 is an illustration of an event **200** being recorded using two mobile devices (i.e. first device **120** and second device **140**). In this embodiment, first user **220** is operating first device **120**, and second user **240** is operating second device **140** (e.g. first and second smartphones). In this example, the event **200** is a live band playing in public. However, this event could be any event witnessed by multiple people with mobile devices. The event may be for example, an exceptional astronomical event (e.g. a large full moon, meteor, or eclipse), an accident (e.g. vehicular), a street performance, or a noteworthy event in a large-scale private gathering such as a company retreat. This system may be particularly valuable in situations where the event is interesting to those who witnessed it, but not noteworthy enough to the broader public to be well documented on the internet.

(9) In FIG. 2, first user **220** is close to the band performance **200**, and is getting relatively good quality video, audio, or photographic recording of band performance **200** on mobile device **120**. On the other hand, second user **240** is across the street from the band (i.e. significantly further away) and records relatively poor quality video, audio, or photographic recording of band performance **200** on mobile device **140**. This results from the fact that mobile devices generally produce lower quality recordings at relatively greater distances and higher quality recordings at relatively shorter distances to the event being recorded.

(10) FIG. 3 shows a comparison between a first recording **320** (i.e. the first media file, or the image recording) recorded on first mobile device **120** and the second recording **340** (i.e. the second media file, or the other image recording) recorded on second mobile device **140**, and computer vision techniques used herein. As shown in FIG. 2, at event **200**, first user **220** records first recording **320** using the camera **128** and/or microphone **127** of first device **120**. First recording **320** is stored on memory **126** of first device **120** in association with first metadata. Also at event **200**, second user **240** records second recording **340** using the camera **148** and/or microphone **147** of second device **140**. Second recording **340** is stored on memory **146** of second device **140** in association with second metadata.

(11) In the example of FIG. 3, first recording **320** is being played back or recorded on first device **120**, and is being displayed on display **129** of first device **120**. Similarly, second recording **340** is being played back or recorded on second device **140**, and is being displayed on display **149** of second device **140**. In this example, first recording **320** is of better quality than second recording **340**, because it was recorded at a closer distance. However, other factors could affect the relative quality of the two recordings. For example, one of the devices could have a superior lens (e.g. long focal length or telephoto), microphone, or recording hardware/software. Alternatively, user settings such as the quality level and recording size could affect relative quality of the recordings.

(12) The software or server **110** determines whether first recording **320** and second recording **340** are of the same event (using processor **112** and memory **116**). In order to do this, server **110** reviews metadata of the first recording **320** and second recording **340** (i.e. first metadata and

second metadata). This metadata is acquired via network **130** and communicator **114**. In some embodiments, this metadata contains location and time information of the recordings. The location information can be obtained using GPS data from the mobile device, or wireless network location tracking. The time information can be obtained from internal clocks in the mobile device, or a time obtained over a network.

(13) Server **110** reviews the metadata of recordings from several different mobile devices to find matches in the metadata indicating recordings of the same event. In one embodiment, the plurality of mobile devices automatically send the metadata to server **110** via network **130**. In other embodiments, this is done only once user approval for the media sharing feature is obtained. In some embodiments, if first recording **320** and second recording **340** have matching location and time metadata, the algorithm considers them a preliminary match for recording the same event. A match in this situation would be overlapping or similar time metadata, and location metadata within a predetermined distance threshold (e.g. 100 meters) of each other.

(14) In one embodiment, the preliminary match based on metadata is established by setting a threshold of similarity between the first recording **320** and the second recording **340**. The server **110** compares the first metadata and the second metadata to determine the similarity between the metadata. This similarity may be calculated based on a weighted algorithm. For example, the similarity score could be proportional to an amount of recording time overlap of the two recordings, and inversely proportional to a distance between locations of the two recordings. Multipliers for these variables could also be used to weigh the scoring of the similarity of the metadata.

(15) Additional factors other than similarity are considered when the server **110** determines the likelihood that the first recording and second recording are of the same event. Another variable that could suggest the same event is recorded in both recordings is that a Bluetooth communication occurred between first device **120** and second device **240**, especially if the communication occurred near the time of the recordings. Bluetooth or other near-field communication between the devices can be performed by communicators **124** and **144** communicating with each other. The presence or absence of a Bluetooth communication between the devices could be added to a weighted calculation, along with the similarity of the metadata to determine the likelihood that the first recording and second recording are of the same event. In other words, server **110** can determine a similarity between the first media file and the second media file based on a comparison between the first metadata and the second metadata, and determine that the first media file and the second media file are likely recordings of the same event when the similarity exceeds a first threshold.

(16) The first threshold can be a similarity score set based on expected scores from a similarity algorithm as described later. The first threshold can be determined based on trial and error, by finding an optimal balance between false positive and false negative matches. Depending on the server resources available, it may be preferable to either raise or lower this threshold to find potential matching recordings for users.

(17) Moreover, whether or not a Bluetooth communication occurred between the first mobile device **120** and second mobile device **140** could be used in place of location information of the recordings to determine a preliminary match. This could be done using known contact tracing techniques. Specifically, several mobile devices would report their Bluetooth (or other near-field communications) with other mobile devices to server **110**. Server **110** would determine a preliminary match in a situation where: 1) first mobile device **120** and second mobile device **140** had a near-field communication; and 2) first mobile device **120** and second mobile device both recorded a media file within a predetermined time range of the near-field communication. The second determination would be made using the first and second metadata.

(18) Once a preliminary match is established, server **110** instructs (via network **130** and communicator **114**) first device **120** and second device **140** to send copies of first recording **320** and second recording **340** to server **110**. These copies can either be in native format, or downgraded

to lower their size and reduce network bandwidth usage. Upon receiving this instruction, first device **120** sends first recording **320** to server **110** (via communicator **124** and network **130**). Similarly, second device **140** sends second recording **340** to server **110** (via communicator **144** and network **130**).

(19) Once server **110** obtains first recording **320** and second recording **340**, server **110** compares them to determine if they are recordings of the same event **200**. To do this, image recognition or computer vision techniques are used to make a final match establishing that the same event was recorded, and to establish which of the two recordings is of higher quality. For example, image recognition can be used to establish that the same objects are located in both recordings. Similarly, sound matching (i.e. hidden Markov model analysis) can be used to establish that the same sounds are in both recordings.

(20) Known image recognition techniques are used to determine if the same event has been recorded in the first recording **320** and the second recording **340**. For example, classification, tagging, object detection, and segmentation can be used. With classification, a class is assigned to the recording, and this narrows the possible matches between the recordings. With tagging, several aspects of the recording are tagged, to provide a more detailed basis for matching the two recordings. With detection, an artificial intelligence (AI) or machine learning algorithm detects specific objects or events within the recording. A bounding box **360** can be placed around objects or events of interest, which the algorithm can focus on in the recording matching process. With segmentation, an element can be located in a recording to the nearest pixel, which can precisely locate objects or events within a recording.

(21) To determine which objects are the subject of the first recording **320**, the algorithm can employ a several techniques, as shown in FIGS. **3** and **4**. For example, the algorithm can focus on portions of the recordings near the center of the recordings for a majority of the recording time. While focusing on objects near the center of the recordings for a majority of the recording time, the algorithm can recognize objects or events of interest using keypoints **400** or bounding boxes **360**. This way, the algorithm can compare the objects or events in first recording **320** and second recording **340** to determine if there is a match. For example, the algorithm can assign keypoints **400** using known techniques on an object or event deemed to be a subject of the first recording **320** (i.e. points of high contrast or unusual brightness). Then, it can search for those keypoints **400** in the second recording **340** to determine if the first recording **320** and second recording **320** are of the same event. For a video recording, the algorithm can compare frames with a same timestamp for instances of the object or event of interest.

(22) In some embodiments, the software trains a deep learning algorithm to recognize the subject of the first recording **320**. In other words, the subject of the first recording **320** is identified (e.g. using its location near the center of the recording) and tagged by the algorithm. This can be done, for example, by using the keypoints **400** and applying nearest-neighbor classification keypoints **400** and other points in the first recording **320**. In particular, this classification and training can be focused near the center of the frames of the first recording **320**. This way, the algorithm can be better trained to recognize the subject of the first recording in other recordings (e.g. the second recording **340**).

(23) The first recording **320** is then used as training data for an image recognition algorithm. The second recording **340** is then inputted into the trained image recognition algorithm to first identify whether the subject of the first recording **320** is located in the recording, and to determine a relative position of the second device **140** relative to the first device **120** when the recordings were made. Of course, the performance of these steps could be reversed. In the embodiments herein, first recording **320** is the superior recording. Thus, the software would ultimately perform its function by identifying first recording **340** using data training from second recording **340**.

(24) In particular, after the keypoints **400** are assigned using first recording **320**, they are located in second recording **340**. Next, the software determines a number of the keypoints **400** that are



locatable on the subject of the second recording **340** (i.e. the second subject). To do this, the algorithm locates an area in second recording **340** where the subject is expected to be located (e.g. the center 25% of the area of one or more frames). The area in of the expected location of the subject can be the same as bounding box **360**, or larger than bounding box **360**. The method for determining the area of the expected second subject can be the same as the method used to determine the location of the first subject.

(25) Once this area is determined, the algorithm attempts to locate keypoints **400** in that area. When the number of locatable keypoints exceeds a threshold, determining that the first subject (subject of the first recording **320**) and the second subject are the same subject (i.e. event **200**). For example, if the image recognition algorithm assigns **100** keypoints to the first subject, and can identify **75** keypoints in an area that the second subject is expected, the threshold is exceeded. Thus, it is determined that the first subject and second subject are the same subject.

(26) In some embodiments, the mobile devices **120**, **140** can locally identify keypoints **400**, bounding boxes **360**, areas of expected location of the subject, and/or a description of the subject using machine learning and computer vision techniques (using processors **122,142**). This information can be added to the metadata. Thus, when the first metadata is sent to server **110**, server **110** can use the metadata to more easily make a preliminary match between first recording **320** and second recording **340** and/or more easily make a final determination of whether the first recording **320** and second recording **340** are the same event. Thus, some of the computational resources of matching the different media files can be transferred from server **110** to mobile devices **120**, **140**. In other words, some of the subject identification and classification is performed by processors **122** and **142**, instead of processor **112**. Thus, a distributed computational approach can be taken, improving the functionality and performance of the network.

(27) Once it is determined that first recording **320** and second recording **340** are of the same event **200**, the algorithm determines which of the first recording **320** and second recording **340** is of better quality. This can be done by determining a relative size of the objects of interest in each recording. For example, if the first recording **320** has a greater distance **402** between the object keypoints **400**, or a larger bounding box **360**, it can be deemed the higher quality recording. The distance **402** between object keypoints **400** may be a pixel distance, which is a number of pixels between the object keypoints **400**. This number of pixels can be an absolute number of pixels, or a number of pixels relative to the resolution of the recording. Alternatively or additionally, the algorithm can review the resolution, bitrate, frame rate, or other recording quality statistics when assessing the relative quality of the recordings.

(28) The algorithm can also determine the relative quality of the first recording **320** and second recording **340** based on lighting conditions. For example, if the algorithm identifies glare and/or poor contrast in one of the recordings, it could determine that the other recording is higher quality. All of the aforementioned factors and other factors can be weighted and tallied in order to generate a quality score for each of the first recording **320** and the second recording **340**. The algorithm determines the higher quality recording based on the recording with the better score.

(29) Once server **110** determines which of the first recording **320** and second recording **340** is of better quality, it sends a link to the device with the lower quality recording that enables access to the higher quality recording. In the embodiment of FIG. 3, first recording **320** is of better quality than second recording **340**. Therefore, server **110** would send a link to first recording **320** to second device **140**. An embodiment illustrating how this link is displayed to the second user **240** is shown in FIG. 5.

(30) FIG. 5 shows a user interface for using the software according to embodiments herein. In FIG. 5, the second user's photo album **500** is open and being viewed as a plurality of tiles of photos and videos on second mobile device **140**. In this embodiment, second mobile device **140** is being used, and second recording **340** of the event is located in the photo album. Second recording **340** (or an icon thereof) is being selected by the second user **240** and appears larger than other media icons in

photo album **500**. Above photo album **500**, a related media pane **502** is displayed. In this media pane, an icon representing first recording **320** is displayed simultaneously with second recording **340** being selected by the second user **240**. Although the software has already determined that first recording **320** is a better (i.e. higher quality) recording than second recording **340** of the event, second user **240** can decide whether or not first recording **320** is desirable for viewing.

(31) In some embodiments, although the icon for first recording **320** is displayed on the second mobile device **140**, the full file is not transferred to that device until the first recording **320** is selected by the second user **240** for viewing. The icon of first recording **320** can be shown as a preview of the recording, either as a still frame or video icon. In some embodiments, the first recording **320** can be streamed by the second user **240** for viewing; in other embodiments, the first recording **320** can be downloaded to second mobile device **140**.

(32) In some embodiments, multiple other recordings of the event recorded in second recording **340** are identified by the software as higher quality than second recording **340**, and multiple icons (i.e. multiple links) corresponding to those recordings are displayed in the related media pane **502**. These multiple higher quality recordings can be from multiple different mobile devices. The software may choose a variety of different recordings of the same event from different vantage points, in order to give the second user **240** many options for viewing the event to select a favorite recording.

(33) Embodiments of the server and mobile device described herein improve the functioning of the computer network by providing desired media to users of the mobile devices. To do this, the various embodiments use analysis techniques limited to the realm of computer technology, such as metadata and machine learning. Moreover, this media is recorded and distributed using computer hardware such as cameras, processors, memory, and network communication hardware. Finally, the problem of poor quality media recordings on mobile devices being addressed herein is a problem limited to the realm of computer technology. Thus, embodiments herein are necessarily rooted in computer technology in order to overcome a problem specifically arising in the realm of computer networks.

(34) As will be appreciated by one skilled in the art, aspects of the present invention may be embodied as a system, method or computer program product. Accordingly, aspects of the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects of the present invention may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(35) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a non-transitory computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the non-transitory computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a non-transitory computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(36) A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave.

Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a non-transitory computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device.

(37) Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing. Computer program code for carrying out operations for aspects of the present invention may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server (e.g. the cloud). In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(38) Aspects of the present invention are described above with reference to flowchart illustrations and block diagrams of methods, apparatuses (systems) and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. Such processor can include processor **112**, processor **122**, and processor **124**.

(39) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flowchart and/or block diagram block or blocks.

(40) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

(41) The flowcharts and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or

flowchart illustration, can be implemented by special purpose hardware-based systems which perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(42) Although the invention has been described with reference to embodiments herein, those embodiments do not limit the scope of the invention. Modifications to those embodiments or different embodiments may fall within the scope of the invention.

#### PART LIST

(43) **100** System **110** Server **112** Processor (Server) **114** Communicator (Server) **116** Memory (Server) **120** First Device **122** Processor (First Device) **124** Communicator (First Device) **126** Memory (First Device) **127** Microphone (First Device) **128** Camera (First Device) **129** Display (First Device) **130** Network **140** Second Device **142** Processor (Second Device) **144** Communicator (Second Device) **146** Memory (Second Device) **147** Microphone (Second Device) **148** Camera (Second Device) **149** Display (Second Device) **200** Event **220** First User **240** Second User **320** First Recording **340** Second Recording **360** Bounding Box **400** Keypoints **402** Keypoint Distance **500** Photo Album **502** Related Media Pane

## Claims

1. A server comprising: at least one memory configured to store instructions; and at least one processor connected to the at least one memory and configured to execute the instructions to: acquire a first media file and first metadata of the first media file; acquire a second media file and second metadata of the second media file; store the first media file, the first metadata, the second media file, and the second metadata in the at least one memory; determine a similarity between the first media file and the second media file based on a comparison between the first metadata and the second metadata; determine that the first media file and the second media file are likely recordings of the same event based on the similarity exceeding a first threshold; and based on a determination that the first media file and the second media file are likely recordings of the same event: confirm that the first media file and the second media file are recordings of the same event by: determining a first subject of the first media file and a second subject of the second media file, by analyzing frames within the first media file and the second media file; and comparing the first subject and the second subject to determine if they are the same subject; determine, based on a comparison between the first media file and the second media file, which of the first media file and the second media file is a higher quality recording of the same event; and based on the second media file being the higher quality recording, send a link to the second media file to a mobile device displaying the first media file, a link to the first media file, or a preview of the first media file.
2. The server of claim 1, wherein the first metadata comprises first location and time information of the first media file, the second metadata comprises second location and time information of the second media file, and the at least one processor is configured to execute the instructions to determine that the similarity exceeds the first threshold based on the first location and time information matching the second location and time information.
3. The server of claim 2, wherein the first location and time information comprises a first recording location and a first recording time of the first media file, the second location and time information comprises a second recording location and a second recording time of the second media file, and the at least one processor is configured to execute the instructions to determine that the first location and time information matches the second location and time information based on the first recording time overlapping the second recording time, and the first recording location being closer than a distance threshold to the second recording location.
4. The server of claim 1, wherein the at least one processor is configured to execute the instructions to: based on the link to the second media file being selected, send the second media file to the mobile device.

5. The server of claim 1, wherein the at least one processor is configured to execute the instructions to compare the first subject and the second subject to determine if they are the same subject by: assigning first keypoints to the first subject; locating second keypoints, corresponding to the first keypoints, on the second subject; and based on a number of the second keypoints exceeding a threshold, determining that the first subject and the second subject are the same subject.
6. The server of claim 5, wherein the at least one processor is configured to execute the instructions to determine which of the first media file and the second media file is the higher quality recording of the same event by: obtaining a first pixel distance between two of the first keypoints; obtaining a second pixel distance between two of the second keypoints corresponding to the two of the first keypoints; and based on the second pixel distance being higher than the first pixel distance, determining that the second media file is the higher quality recording.
7. The server of claim 1, wherein the at least one processor is configured to execute the instructions to determine which of the first media file and the second media file is the higher quality recording of the same event by: assigning a first bounding box to the first subject; assigning a second bounding box to the second subject; and based on the second bounding box being larger than the first bounding box, determining that the second media file is the higher quality recording.
8. The server of claim 1, wherein the at least one processor is configured to execute the instructions to analyze first center regions of the frames within the first media file and second center regions of the frames within the second media file to determine the first subject and the second subject, respectively, by: identifying at least one first object in the first center regions; determining that the at least one first object is the first subject if it is in the first center regions for a majority of a recording time of the first media file; identifying at least one second object in the second center regions; and determining that the at least one second object is the second subject if it is in the second center regions for a majority of a recording time of the second media file.
9. The server of claim 1, wherein the at least one processor is configured to execute the instructions to, after acquiring the first media file and the second media file, confirm that the first media file and the second media file are recordings of the same event by: comparing audio recordings in the first media file and the second media file at corresponding time points; and based on a similarity in the audio recordings exceeding a threshold, confirming that the first media file and the second media file are recordings of the same event.
10. The server of claim 1, wherein the first metadata comprises a description of the first subject, and the second metadata comprises a description of the second subject, and the at least one processor is configured to execute the instructions to determine that the similarity exceeds the first threshold based on the description of the first subject and the description of the second subject matching.
11. A device comprising: at least one memory configured to store: instructions; a first media file; first metadata of the first media file; a second media file; and second metadata of the second media file; a display; and at least one processor connected to the at least one memory and the display, the at least one processor being configured to execute the instructions to: determine a similarity between the first media file and the second media file based on a comparison between the first metadata and the second metadata; determine that the first media file and the second media file are likely recordings of the same event based on the similarity exceeding a first threshold; and based on a determination that the first media file and the second media file are likely recordings of the same event: confirm that the first media file and the second media file are recordings of the same event by: determining a first subject of the first media file and a second subject of the second media file, by analyzing frames within the first media file and the second media file; and comparing the first subject and the second subject to determine if they are the same subject; determine, based on a comparison between the first media file and the second media file, which of the first media file and the second media file is a higher quality recording of the same event; based on the first media file being the higher quality recording, instruct the display to display a link to the first media file; and

based on the second media file being the higher quality recording, instruct the display to display a link to the second media file.

12. The device of claim 11, wherein the first metadata comprises first location and time information of the first media file, the second metadata comprises second location and time information of the second media file, and the at least one processor is configured to execute the instructions to determine that the similarity exceeds the first threshold based on the first location and time information matching the second location and time information.

13. The device of claim 12, wherein the first location and time information comprises a first recording location and a first recording time of the first media file, the second location and time information comprises a second recording location and a second recording time of the second media file, and the at least one processor is configured to execute the instructions to determine that the first location and time information matches the second location and time information based on the first recording time and the second recording time overlapping, and the first recording location being closer than a distance threshold to the second recording location.

14. The device of claim 11, wherein the at least one processor is configured to execute the instructions to: instruct the display to display a preview of the first media file or the second media file.

15. The device of claim 11, wherein the at least one processor is configured to execute the instructions to compare the first subject and the second subject to determine if they are the same subject by: assigning first keypoints to the first subject; locating second keypoints, corresponding to the first keypoints, on the second subject; and based on a number of the second keypoints exceeding a threshold, determining that the first subject and the second subject are the same subject.

16. The device of claim 15, wherein the at least one processor is configured to execute the instructions to determine which of the first media file and the second media file is the higher quality recording of the same event by: obtaining a first pixel distance between two of the first keypoints; obtaining a second pixel distance between two of the second keypoints corresponding to the two of the first keypoints; and based on the second pixel distance being higher than the first pixel distance, determining that the second media file is the higher quality recording.

17. The device of claim 11, wherein the at least one processor is configured to execute the instructions to determine which of the first media file and the second media file is the higher quality recording of the same event by: assigning a first bounding box to the first subject; assigning a second bounding box to the second subject; based on the first bounding box being larger than the second bounding box, determining that the first media file is the higher quality recording; and based on the second bounding box being larger than the first bounding box, determining that the second media file is the higher quality recording.

18. The device of claim 11, wherein the at least one processor is configured to execute the instructions to analyze first center regions of the frames within the first media file and second center regions of the frames within the second media file to determine the first subject and the second subject, respectively, by: identifying at least one first object in the first center regions; determining that the at least one first object is the first subject if it is in the first center regions for a majority of a recording time of the first media file; identifying at least one second object in the second center regions; and determining that the at least one second object is the second subject if it is in the second center regions for a majority of a recording time of the second media file.

19. The device of claim 11, wherein the at least one processor is configured to execute the instructions to confirm that the first media file and the second media file are recordings of the same event by: comparing audio recordings in the first media file and the second media file at corresponding time points; and based on a similarity in the audio recordings exceeding a threshold, confirming that the first media file and the second media file are recordings of the same event.

20. The device of claim 11, wherein the first metadata comprises a description of the first subject,

and the second metadata comprises a description of the second subject, and the at least one processor is configured to execute the instructions to determine that the similarity exceeds the first threshold based on the description of the first subject and the description of the second subject matching.

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